



Complex Systems

2021/2022

Exam R 08 of July 2022 Time: 2h00

Name:

Number:

This is a **closed books** exam. Any attempt at cheating implies immediate disapproval in the course and the activation of the procedures defined in the disciplinary regulation document of the UC.

Question #	Points	Your score
1	18	
2	12	
3	25	
4	25	
5	20	
	100	

Grader:

Question 1 - Define 18 points

Define the following concepts in a clear and **rigorous** way. You may use a concrete example to **help you** to explain the concept.

a) Basin of Attraction.

Your answer:

b) Cellular Automata.

Your answer:

c) Bipartite Graph.

Your answer:

Question 2 - Comment 12 points

Comment the following statements. **Remember**, to comment is not just to say agree/disagree for you have to justify your evaluation.

- a) The logistic equation in the chaotic regime is statistically predictable even if the orbits are unpredictable.

Your answer:

Your answer:

- b) In terms of robustness it is better to have a system modeled as a small world rather than system whose model is a scale free network.

Your answer:

Your answer:

Question 3 - [Simulate](#) 25 points

Suppose you have a model of a system described by the differential equation $\frac{dx}{dt} = f(x)$. The only thing you know about $f(x)$ is given by the plot of figure 1.

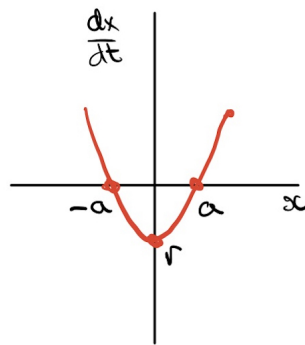


Figure 1: My system

Based on this information what is a possible form of $f(x)$? Moreover, does your system have fixed-points? If so, identify them including their nature.

Your answer:

Your answer:

Question 4 - Explain 25 points

The **Julia Set** was discussed in class. Describe its meaning and give a formal definition of it. How can we obtain a Julia Set using an Iterated Function System (IFS)?

Your answer:

Your answer:

Question 5 - [Model](#) 20 points

In an island the population is divided into two groups: those that love dolphins and those that hate dolphins. Every day each person talks to each other ~~person~~ exactly once about the dolphins. There is a certain chance that a lover becomes a hater and vice-versa. Using **difference equations model the problem** of the time evolution of lovers and haters. Make clear your **assumptions** and discuss the **dynamics** of your system.

Your answer:

Your answer: